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Tailoring electrochemical properties of graphene oxide via quinone-Nitrogen functionalization for high energy supercapacitor

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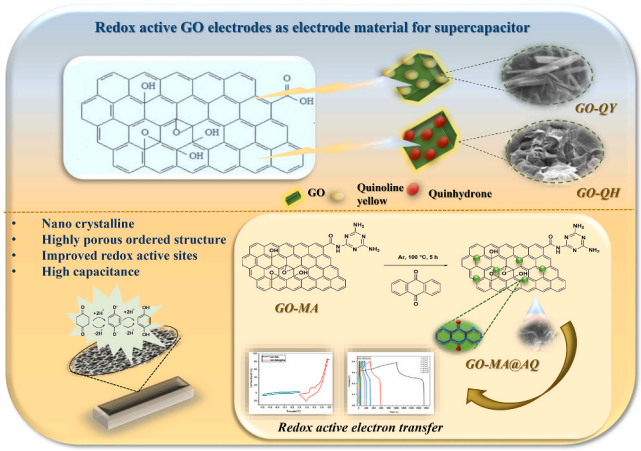
HIGHLIGHTS

- Graphene oxide (GO) nanosheets were functionalized with quinone-N derivatives via a thermo-sonochemical route.
- The synergistic effect of multiple N-configuration along with quinone groups boosts the overall electrochemical performance.
- GO-QY and GO-MA@AQ showed exemplary electrochemical performance, with E_d of 85 and 148 W h Kg^{-1}, respectively.
- This study presents the electrochemical behavior of quinone-N moieties in GO, aiding design of advanced electrode materials.

GRAPHICAL ABSTRACT



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ABSTRACT

Organic supercapacitors are gaining significant interest due to their impressive pseudocapacitive behavior, making them ideal for high-end energy storage devices. Herein, electroactive graphene oxide electrodes using quinone-Nitrogen derivatives are synthesized via a thermo-sonochemical process. These electrodes have outstanding electrochemical performance due to their relatively low band gap energies. The electrical conductivity of the synthesized GO electrodes demonstrated high capacitance, significantly higher than that of the reference material, graphene oxide. Among all the synthesized quinone derivatives, GO-QY and GO-MA@AQ showed commendable electrochemical performance, with exemplary electrode capacitances of 1052 F g^{-1} and 1853 F g^{-1}, respectively. Correspondingly, they achieved energy densities of 85 and 148 W h kg^{-1} with a maximum power density of 9.4 and 14.4 W kg^{-1}. Notably, GO-MA@AQ retained 90 % of its initial capacitance after 5000 continuous cycles. The superior capacitive behavior of these functionalized electrode materials is attributed to the reversible redox functions of the nitrogen functionalities and quinone groups, which enhance the overall charge storage performance. Although melamine and quinoline yellow have been widely studied for applications in polymer chemistry and biosensing, respectively, their electrochemical roles in supercapacitor systems remain largely unexplored due to limitations such as low conductivity and poor cycling stability. The

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findings from this study indicate that quinone-N functionalized graphene oxide has great electrochemical potential as a hybrid electrode material in high-performance supercapacitors.

1. Introduction

The global shift toward sustainable energy solutions has intensified the demand for advanced materials with superior electrochemical performance, particularly for high-performance supercapacitors. These energy storage devices offer high power density, rapid charge-discharge rates, and long cycle life, making them ideal for applications ranging from portable electronics to electric vehicles [1]. To meet growing energy demands, the scientific community is actively engaged in the design and development of novel electrode materials that can deliver enhanced energy density without compromising cycling stability. Recent advancements in supercapacitor technology have focused on integrating redox-active materials to achieve high energy density without compromising power output [2]. The charge storage capacity of these supercapacitors can be significantly enhanced through the selection of materials that facilitate surface redox reactions [3]. Their electrochemical performance is further exemplified by the synergistic interaction between electrode and electrolyte materials [4,5]. In this regard, Organic electrode materials have gained significant attention in recent years due to their unique advantages, such as structural tunability, low environmental impact, and lightweight nature. Their intrinsic redox-active sites enable promising pseudocapacitive behaviour [6]. However, their practical implementation is often constrained by poor electrical conductivity, mechanical instability, and solubility issues in electrolytes [7]. These limitations have led to the integration of organic moieties with highly conductive supports, such as metal oxides, graphene, and carbon nanotubes, to fabricate hybrid electrodes with improved performance [8]. Metal oxides are a diverse class of materials that have found widespread applications across multiple scientific domains due to their unique physical, chemical, and electronic properties. They possess high thermal and chemical stability, and tunable band gaps, which make them versatile for a range of functional uses [9]. In the field of energy storage, transition metal oxides such as MnO_2, Co_3O_4, NiO, and Fe_3O_4 are of particular interest due to their high theoretical capacitance, multivalent redox properties, and ability to store charge through faradaic processes [10]. Despite their advantages, challenges such as poor electrical conductivity and structural degradation during cycling exist. However, these are increasingly addressed by forming composites with conductive carbon materials or by nanostructuring the oxides to enhance ion diffusion and electron transport. Hybrid electrodes combining transition metal oxides with carbon supports have shown energy densities exceeding 60 Wh kg^{-1} [11].

Among carbon-based materials, Graphene oxide (GO), owing to its tunable surface chemistry and large specific surface area, has emerged as a promising scaffold for electrochemical energy storage. The presence of oxygenated groups in GO disrupts the pristine sp^2 hybridized carbon network of graphene, introducing defects and increasing the interlayer spacing, which can be advantageous for ion accessibility and charge storage. The performance of graphene-based electrodes in supercapacitors is inextricably linked to their energy density, a critical parameter that dictates the amount of energy a device can store [12]. In comparison, reduced graphene oxide (rGO) obtained by the partial removal of oxygen functionalities from GO exhibits improved electrical conductivity and structural integrity [13], and has been extensively employed in various electrochemical applications, including batteries, sensors, and supercapacitors. Despite its conductivity advantage, rGO tends to exhibit lower surface functional group density, which limits its capacity for pseudocapacitive interactions [14]. Conversely, emerging evidence suggests that, compared to rGO, GO offers greater chemical tunability. However, its practical performance is often hindered by limited pseudocapacitive behavior, primarily due to the presence of insufficient redox-active sites. Notably, the abundant oxygen-containing functional groups in GO can act as anchoring sites for redox-active molecules, thereby enhancing its suitability for designing organic-inorganic hybrid electrode. Additionally, the larger interlayer spacing in GO enhances electrolyte ion diffusion, a feature that is often compromised in compact rGO sheets. By integrating redox-active organic components with graphene oxide through robust chemical interactions, it is possible to enhance electron transport, introduce multiple charge storage mechanisms, and improve structural stability during prolonged cycling. Organic molecules, particularly those bearing quinone functionalities [15], offer pronounced redox activity and tunable electronic properties, making them ideal candidates for functionalizing GO to achieve high-performance energy storage. Despite their promising redox characteristics, quinone-based materials often suffer from limited surface area and suboptimal electronic conductivity, which constrain both their charge storage capacity and rate performance. However, a comprehensive understanding of how the chemical structure of quinone can influence the electrochemical behavior in such hybrid systems remains underexplored. This study addresses this gap by synthesizing a novel organic-GO hybrid material, wherein quinone [16] and nitrogen-rich organic unit, melamine, are anchored onto GO surfaces to simultaneously improve conductivity, redox activity, and structural integrity. Our approach not only overcomes the limitations of pristine GO but also leverages the advantages of organic molecules to deliver high capacitance and outstanding cyclic performance.

Recent studies have highlighted the significance of organic electronics that are primarily governed by redox-active carbonyl functionalities and radical polymers potentially applied in capacitors and electrochromic devices [17]. Functionalizing graphene oxide with organic molecules can enhance its supercapacitor performance by introducing redox-active species or proton-conducting moieties. Numerous redox-active organic materials have been studied in energy storage systems in which the organic redox functions (quinone, imine (C═N), triazine(C–N)) are of high interest because of their exemplary redox performance [18]. The use of quinone moieties in graphene-based materials significantly improves their electrochemical performance as the redox activity of quinones contributes to pseudocapacitance, increasing the overall charge storage capacity of the material. However, further enhancement in the structure, along with increased surface area of the electrode materials is still needed in the research of quinone-modified graphene oxide. Thus, heteroatom incorporation in graphitic carbon materials [19] has emerged as a promising strategy for imposing structural modification, which can further tune their electronic structure, thus resulting in enhanced electrochemical performances in the field of energy storage devices. The incorporation of nitrogen atoms in the graphitic structure enhances the electronic conductivity by facilitating faradaic reactions on the nitrogen-mediated active sites, during the electrochemical charge-discharge process [20] resulting in high capacitance. Upon incorporation into the graphene framework, nitrogen atoms adopt distinct configurations, primarily as graphitic N, pyridinic N, and pyrrolic N. Each of these configurations modulates the electrochemical properties of the material that are beneficial for supercapacitor applications. Graphitic N, where the nitrogen atom substitutes a carbon atom within the basal plane, contributes to enhanced electrical conductivity by increasing electron density in the delocalized π-system. In contrast, pyridinic and pyrrolic N, being embedded in the graphene plane, introduce electron-rich active sites that facilitate faradaic redox reactions, thereby boosting pseudocapacitive performance. These nitrogen functionalities significantly enhance electrode wettability and amplify pseudocapacitive contributions. For example, Fan et al. [21] synthesized Nitrogen-doped graphene hollow

spheres (NGHS) using a simple layer-by-layer assembly exhibiting a maximum capacitance of 381 F g^{-1} at a current density of 1 A g^{-1}. Dai et al. [22] prepared N-doped graphene with 3D porous architecture with a high specific capacitance of 390 F g^{-1}, wherein the fabricated symmetric SC device retained 96 % of its initial capacitance after 20,000 cycles. In a recent work, nitrogen-doped graphene electrodes for high-performance supercapacitors has been demonstrated using Yucel's method [23]. The specific capacitance of the electrodes changed from 178 mF cm^{-2} to 2034 mF cm^{-2} in 10 mA cm^{-2} current density as a function of the mesoporous structure. Among various heteroatom rich organic molecules, Melamine, a nitrogen-rich compound (66 % nitrogen by mass) [24], is chosen for introducing nitrogen functionalities, which contribute to pseudocapacitive behavior. Incorporating melamine-derived materials not only contributes to improved thermal stability but also significantly extends the cycle life of supercapacitors, thereby rendering them highly suitable for long-duration energy storage applications [25]. The integration of these compounds into conductive graphene oxide creates a synergistic effect by enriching the density of redox-active sites [26], thereby enhancing their electrical conductivity by facilitating charge transfer and modulating band structures.

Addressing the need for high-performance electrode materials, this work presents the synthesis of electroactive organic compounds functionalized onto graphene oxide to investigate the effect of quinone-N structures. Through solvothermal-assisted reaction, all the quinone derivatives are adsorbed onto the surface of GO nanosheets via π-π interactions. Also, to improve the portion of pseudocapacitive contribution, N-rich active centers are introduced into GO using melamine, followed by the anchoring of the reversible electrochemical redox couple, anthraquinone. This strategy facilitates a rapid and reversible $2e^-/2H^+$ redox process of quinone moieties, while the carbon matrix acts as a stabilizing support, effectively mitigating quinone degradation during electrochemical cycling. The electrochemical performances are determined from cyclic voltammetry and galvanostatic charge-discharge analysis using the as-produced modified GO electrodes. The synthesized materials are evaluated for their charge storage behaviour, and this combination of quinone-Nitrogen groups can deliver increased pseudocapacitive performance, thereby serving as potential electrode materials for supercapacitor applications.

2. Experimental section

2.1. Materials required

All chemical materials were purchased from commercial sources and used without further purification. Graphene oxide (99 % purity) with an average particle size of <10 μm was sourced from Adnano Technologies Pvt. Ltd. Melamine (99.9 % pure), Quinoline yellow, Quinhydrone, and Anthraquinone were purchased from Sigma Aldrich.

2.2. Synthesis of GO-QY and GO-QH

Initially, 500 mg of graphene oxide was dispersed in 150 ml of deionized water into which 50 mg of NaOH solution was added. The resulting mixture was ultrasonicated for 1 h followed by the addition of 250 mg of quinoline yellow. Subsequently, the resulting mixture was sonicated for continuous 5 h. The obtained GO-QY is centrifuged, and the residue is washed with ethanol, dried in a hot air oven for 24 h at 70 °C. In a typical synthesis procedure, 500 mg of graphene oxide was dispersed in 200 ml deionized water. Meanwhile, 250 mg of quinhydrone was dispersed in ethanol separately, and the resulting mixture was added to the above GO suspension and refluxed under argon atmosphere for 12 h. The reaction mixture is filtered, and the residue is washed with ethanol, dried using acetone, followed by vacuum drying at 70 °C to obtain GO-QH.

2.3. Synthesis of GO-MA and GO-MA@AQ

Melamine functionalized graphene oxide (GO-MA) was synthesized by a procedure reported by Parisa et al. [27], with further modifications. Into a mixture of 7.5 ml thionyl chloride in 15 ml of DMF, 2.5 g of graphene oxide was added. The suspension was heated up to 70 °C while being stirred and then was maintained for 20 h under reflux conditions. To this, a mixture of 3 g of melamine and 1 ml of pyridine in 60 mL of DMF was added, and the reaction mixture was stirred at 120 °C for 72 h under reflux conditions. Then, the mixture was cooled down to room temperature and centrifuged and washed thrice with DI water, and dried at 70 °C for 24 h under vacuum. For the synthesis of GO-MA@AQ, 200 mg of GO-MA was dispersed in 100 ml DI water via sonication, into which 175 mg of anthraquinone was added and gently stirred and reflux reaction was carried out under argon atmosphere for 5 h. The above solution was washed with H_2SO_4, and then the colloidal solution was centrifuged (2000 rpm) and washed several times with distilled water, followed by vacuum drying at 70 °C for 24 h.

2.4. Electrochemical measurements

Fabrication of electrodes for electrochemical measurements: The working electrodes are prepared by mixing the GO materials, carbon black, and polyvinylidene fluoride (PVDF) in a mass ratio of 8:1:1 and dispersing them in N-methylpyrrolidone (NMP) as a solvent to form a uniform slurry. The mixtures were then coated onto nickel foam current collectors (0.2 mm thick, 1 cm^2 area) to fabricate electrodes. For all Galvanostatic charge-discharge (GCD) measurements, the active material was uniformly coated onto the current collector with a precise mass loading of 2 mg, ensuring reliable comparison of electrochemical performance across samples. Electrochemical characterization was performed in a conventional three-electrode configuration. The working electrode (Ni foam coated with active material) was tested in a 6 M KOH aqueous electrolyte, with a platinum electrode serving as the counter and an Ag/AgCl (1.0 M KCl) electrode as the reference (Scheme 1).

2.5. Material characterization

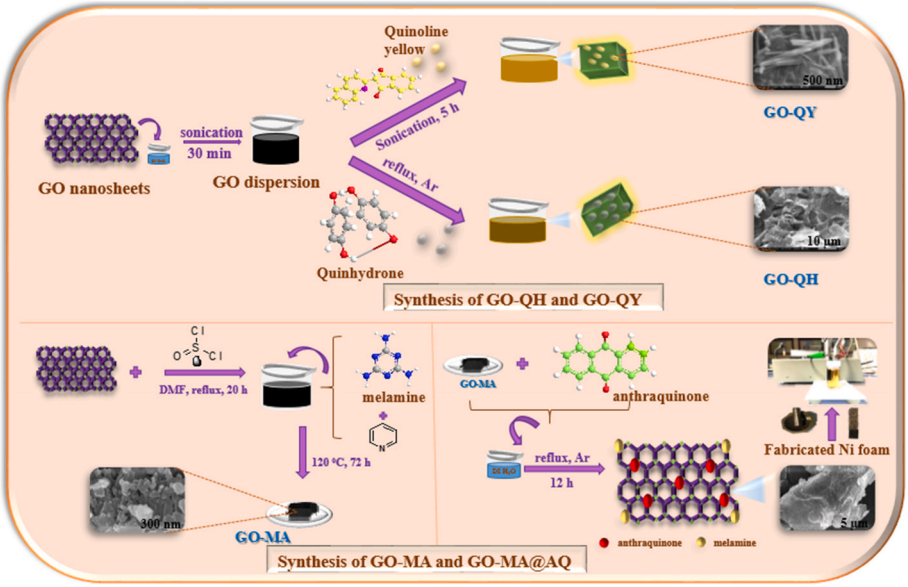
XRD analysis was performed to investigate the crystalline nature and interlayer distance of the synthesized materials using the instrument PANalytical X'Pert Pro in the 2θ range of 10°–90°. JEOL-JSM-IT-200 – IT 200 (SEM) and Carl Zeiss – SIGMA 300 (FESEM) analysis was used to determine the morphological features of the synthesized graphene oxide derivatives. Raman spectroscopy was performed to study molecular vibrations, providing information about structural distortion using WITEC ALPHA 300R. UV-DRS analysis was performed to analyze the optical properties of synthesized photoactive materials using a UV-VIS-NIR Spectrophotometer - Lambda 750. Fluorescent spectral acquisition was carried out to study the emission properties of the compounds using HORIBA Fluorolog-NIR spectrometer.

3. Results and discussion

3.1. Structural analysis

3.1.1. X-ray diffraction studies

X-ray diffraction patterns for all the synthesized samples are summarized in Fig. S1. The XRD pattern corresponding to GO is well matched with the literature [28]. GO-QH (Fig. S1 (a)) revealed intense diffraction peaks at 2θ = 37.35° with an interlayer distance of 1.26 nm. Additional strong and sharp peaks at 43.01° and 45.60°, having a d value of 1.12 and 1.07 nm, respectively, indicate the formation of the target compound. XRD peaks of GO-QY displayed a clear peak at 32.15°, with a d value of 1.44 nm. Peaks at 45.90°, 56.76°, and 75.54° with d-spacing of 1.07, 0.92, and 0.01 nm, respectively, indicate the reduction of the interlayer distance in GO sheets. Fig. S1 (b) depicts the crystalline nature



Scheme 1. Schematic illustration for the preparation of GO electrodes.

of GO-QY. The results obtained from the SEM analysis also indicate its distinctly ordered crystalline morphology. XRD spectrum of GO-MA (Fig. S1 (c)) revealed peaks with significant peak broadening corresponding to melamine. GO-MA shows a non-amorphous and crystalline phase 2θ at 10.4°, 13.22°, 19.3°, 21.74°, 23.3°, 25.28°, 28.9°, and 35.91° [29]. The diffraction peak detected at 2θ = 10.4° is attributed to the (001) plane as indexed for the hexagonal crystalline structure of graphene oxide. Functionalization led to a significant reduction in the interlayer spacing of graphene sheets, decreasing from 0.85 nm (10.4°) to 0.21 nm (43.01°) [30]. With the incorporation of anthraquinone onto GO-MA, distinct peaks at 2θ = 11.37° and 14.20° corresponding to graphene oxide with a significant decrease in the d value (0.39 nm) were obtained (Fig. S1 (d)). The main peak of GO was shifted to the left at 9° with an enlarged spacing of 0.8 nm. However, the additional peaks at 2θ = 23.17°, 25.07°, and 26.37° correspond to the grafting of anthraquinone moieties onto GO-MA, depicting the efficient functionalization (Table 1).

3.1.2. Raman spectroscopy

The structural features of all samples were further analyzed using Raman spectra and is presented in Fig. S2. The Raman spectrum of GO-QH reveals structural features having G band at 1583 cm^{-1} and D band at 1341 cm^{-1}, with an I_D/I_G ratio of 1.01. On the other hand, strong D band at 1350 cm^{-1} and G band at 1594 cm^{-1} have been observed for GO-QY with a I_D/I_G ratio of 0.6 [31]. This signifies functionalization with minimal disruption of the π - conjugated structure rendering a more ordered sp^2 framework. GO-MA represented D and G bands with a significant increase in the intensity of the peaks at 1349 cm^{-1} & 1580 cm^{-1} with an I_D/I_G ratio of 1.02, indicating more crystallinity and defects in the functionalized graphene oxide [32] due to melamine functional groups. Upon anchoring anthraquinone to the melamine-functionalized GO (GO-MA@AQ), a further increase in disorder is observed, with the G band retained at 1583 cm^{-1} and an I_D/I_G ratio rising to 1.16. This pronounced increase is attributed to the synergistic effect of nitrogen inclusion and quinonoid moieties, which further disrupt the sp^2 network, creating a higher density of defect sites. These structural perturbations likely facilitate enhanced pseudocapacitive behavior by promoting accessible redox-active centers. Collectively, the Raman analysis confirms that post-functionalization, the graphitic framework undergoes controlled structural disruption, which is essential for enhancing surface area, defect-driven redox activity, and electrochemical functionality in energy storage applications.

Table 1
The calculated XRD parameters for GO and the synthesized samples.

Sample ID	2θ°	d (nm)
GO	11.21	0.78
GO-QH	37.35	1.26
GO-QY	32.15	1.44
GO-MA	23.3	0.21
GO-MA@AQ	26.37	0.30

3.2. Surface morphology

The micro structures of all GO functionalized compounds were investigated using scanning electron microscopy and field emission scanning electron microscopy. As shown in Fig. 1 (a) (b), GO-QH shows crumpled, flaky layered structure with sharp edges at 2 µm. While GO-QY (Fig. 1 (c) (d)) exhibited nanorods at 500 nm, depicting the highly crystalline morphology. Also, voids and spaces are found at 5 µm with an agglomerated surface, which will enhance the ion mobility, giving more access for ions to interact with the electrode.

Fig. 1 (e) (f) reveals the surface morphology of GO-MA, which exhibits a layered structure, indicating a nano ordering. However, a very thin, silky, paper-like morphology with enhanced transparency was observed for GO-MA@AQ (Fig. 1 (g) (h)) at 2 µm. This suggests a well-ordered carbon structure with uniform distribution of the synthesized material. These structural changes further support the efficient functionalization of graphene sheets, indicating the material's suitability in energy storage applications.

3.3. Optical studies

3.3.1. UV- Diffuse Reflectance Spectroscopy

The optical band gap of the synthesized GO electrodes was determined using UV-Diffuse Reflectance Spectroscopy (UV-DRS), followed by Tauc plot analysis. All the GO electrodes exhibited a narrower band gap as shown in Fig. 2. GO-QH shows a band gap energy of 1.36 eV, while GO-QY and GO-MA@AQ have band gap of 1.45 and 1.43 eV, respectively. However, the band gap energy of the GO electrodes has been remarkably reduced as compared to pure graphene oxide (E_g = 2–4 eV) [33], which is due to the formation of hybrid energy levels,

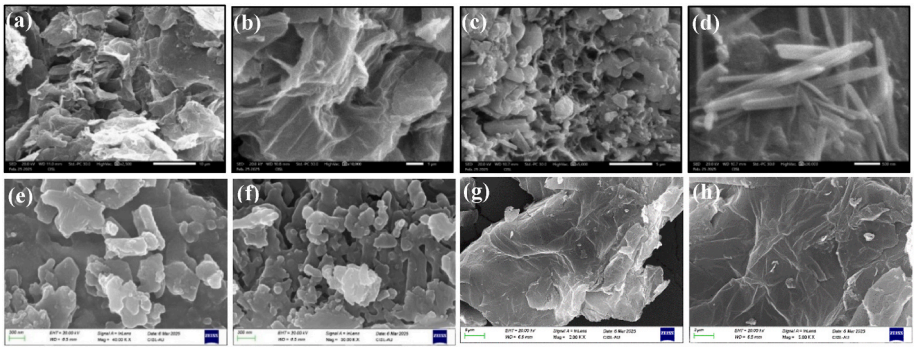


Fig. 1. Morphological analysis of (a & b) GO-QH (c & d) GO-QY; HRTEM images of (e & f) GO-MA (g & h) GO-MA@AQ at low and high magnifications.

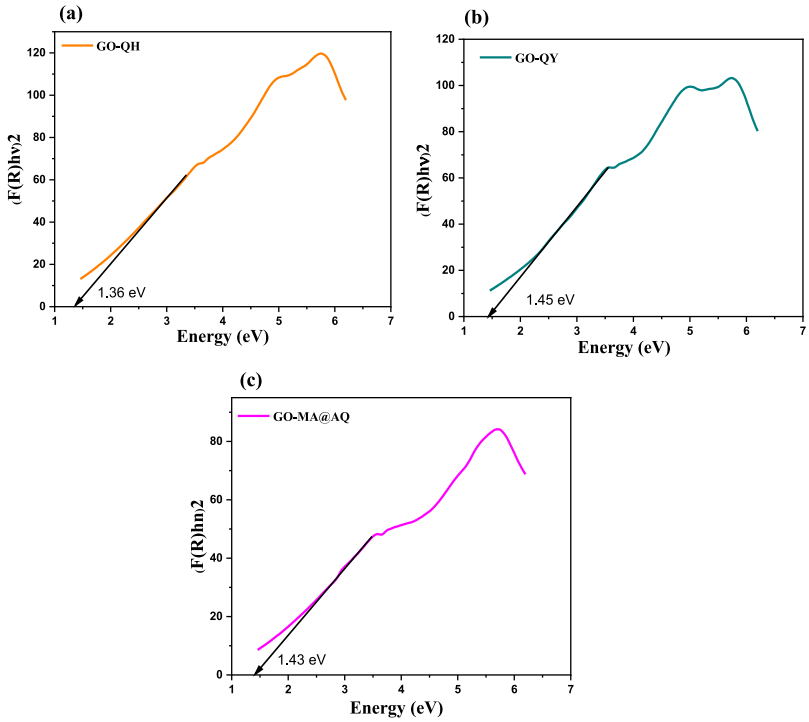


Fig. 2. Tauc plot of (a) GO-QH (b) GO-QY (c) GO-MA@AQ

suggesting the enhanced electronic delocalization and π-π interaction within the molecular framework. This indicates its performance for increased electron transport and rapid charge storage with a very high specific capacitance. From these band gap energies, the radiative transitions occurring within the system is observed from the photoluminescence spectrum (Fig. 3).

3.3.2. Photoluminescence spectroscopy

Photoluminescence spectroscopy serves as a pivotal analytical technique offering profound insights into the electronic structure, the presence of defect states, and the nature of band gap transitions within synthesized materials. In the context of this study, photoluminescence analysis was performed across a spectral range 200–800 nm to comprehensively assess the optical behavior of a functionalized graphene oxide nanocomposite. The optical absorption characteristics of the nanocomposite revealed a pronounced light response within the visible region of the electromagnetic spectrum. The photoluminescence spectra obtained for GO–QH, GO–QY, and GO–MA@AQ distinctly exhibited emission peaks at wavelengths of 433 nm, 536 nm, and 513 nm, respectively. Both GO–QY and GO–MA@AQ relate deep-level recombination centers possibly associated with residual oxygen functional groups and π-conjugation domains. This suggests the formation of defect-induced energy states arising from structural perturbations within the nanomaterial. These defect states are intricately linked to the presence of sp^2 hybridized carbon atoms [34] and the three prominent functional groups inherent to graphene oxide, namely C–O, C═O, and COOH. The observed emissions lend credence to the presence of defective surface characteristics, thereby corroborating the existence of electronic states facilitated by extended π-conjugation and heteroatom functionalization [35].

3.4. Electrochemical analysis

3.4.1. Cyclic voltammetry

To investigate the electroactive properties of the modified GO electrodes, cyclic voltammetry (CV) analysis was carried out using the standard three-electrode setup in a 6 M KOH solution as the aqueous electrolyte. Their electroactivity and charge storage behaviour with redox active sites were assessed. All GO electrodes displayed pseudocapacitive properties, depicting broad redox peaks that are

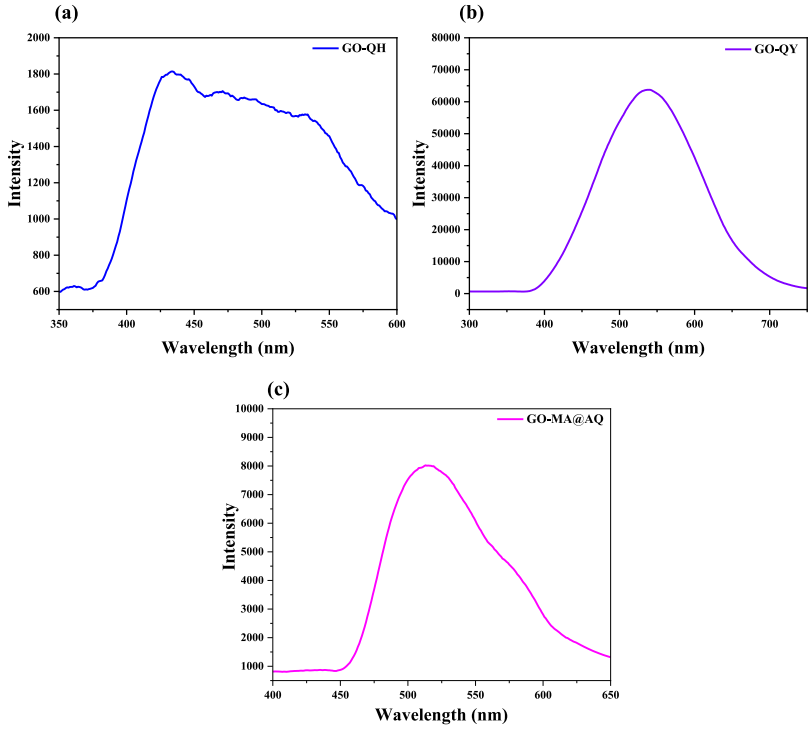


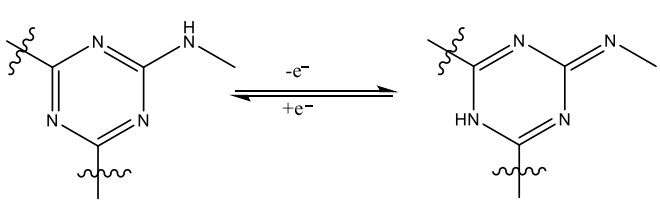
Fig. 3. Photoluminescence spectra of (a) GO-QH (b) GO-QY (c) GO-MA@AQ

characteristic of a charge storage behaviour involving both diffusion and surface-controlled process. The CV curves for all samples were recorded at varying scan rates from 5 mV s^{-1} to 100 mV s^{-1} (Scheme 2).

The comparative CV results in the increasing oxidation and reduction pattern for the synthesized nanomaterials is depicted in Fig. 4. It is evident from Fig. 4 (a) that both GO-QH and GO-QY exhibited faradaic redox reactions below 0.4 V. This can be attributed to the synergistic effect between graphene oxide and redox-active quinone system leading to the two electron reduction and oxidation reaction (Scheme 3) [36]. GO-MA demonstrates minimal redox activity as depicted in Fig. 4 (b). While melamine itself is non-electroactive, the modification of graphene oxide with melamine can enhance the electroactivity of the material, which facilitates rapid faradaic reactions [37]. This confirms the charge storage capability that arises due to nitrogen functionalities contributing to enhanced electron transfer with GO sheets [38].

The further improvement of faradaic pseudocapacitive behaviour with the functionalization of anthraquinone (GO-MA@AQ), as seen in Fig. 4 (b) is attributed to the increased N rich sites due to melamine functionalization and possible electron transfer reaction from AQ, which improves the wettability between the electrodes and electrolytes, consequently improving the whole electrochemical performance. The charge storage of GO-MA@AQ mainly occurred by electrical double layer capacitive (EDLC) mechanism.

The effect of scan rate of the modified GO electrodes was also investigated. Fig. S3 shows the strong dependence of the current with respect to the scan rate. With increase in scan rate, the peak current



Scheme 2. Redox reaction of melamine unit in GO-MA & GO-MA@AQ.

position shifts as the voltage sweep is increased, indicating a quasi-reversible system. This indicates that, in addition to surface-limited processes, the system exhibits redox reactions occurring at the surface of the electrode. The CV curves are more pronounced at lower scan rates, indicating superior redox activity whereas at higher scan rates, the potential changes rapidly, and driving stronger electrochemical response with significant peak broadening. This can be attributed to the increased electrical conductivity of the GO electrodes rendering more pronounced electrochemically accessible surface area with increasing scan rates. Additionally, as the scan rate increases, anodic and cathodic peaks shift to higher and lower current density regions, a behaviour primarily attributed to the even distribution of electrolyte ions into the electrode's surface thereby resulting in enhanced capacitive behaviour [39].

From the cyclic voltammetric results, anodic and cathodic peak currents with normalized scan rate were plotted for GO-QY & GO-MA@AQ. This exhibits a linear relation ($i \alpha \nu$) which is characteristic of a capacitive process Fig. S4 (a) & (c) [40]. Thus, in order to assess the charge storage kinetics and to elucidate the contribution from EDLC and pseudocapacitance, ($i \alpha \nu^{1/2}$), the two processes can be expressed mathematically by the following relation, $i(\nu) = k_1\nu + k_2\nu^{1/2}$. The peak currents for both anodic and cathodic peak potentials (I_{ap} & I_{cp}) vs $\nu^{1/2}$ vary linearly as shown in Fig. S4 (b & d). This suggests that both GO-QY & GO-MA@AQ have also participated in a quasi-reversible diffusion-controlled process. Thus, the contribution from capacitive (EDLC) and diffusion-controlled processes was quantified from the Dunn method. Fig. S5 (a) and (b) reveal the quantification of charge contribution. At lower scan rates, for GO-QY (Fig. S5 (a)) diffusive contribution dominates, reflecting faradaic redox reactions with faster kinetics. At higher scan rate, both diffusive and capacitance contributions are almost the same, proposing a balanced quasi-reversible redox mechanism. But for GO-MA@AQ (Fig. S5 (b)), the graphical plot clearly indicates that the charge storage mechanism is predominantly governed by diffusion-limited process [41]. This is further evidenced by the enhanced capacitive response obtained at higher scan rates, where the contribution from diffusion-mediated process becomes increasingly marginal, highlighting the dominance of rapid surface-mediated redox

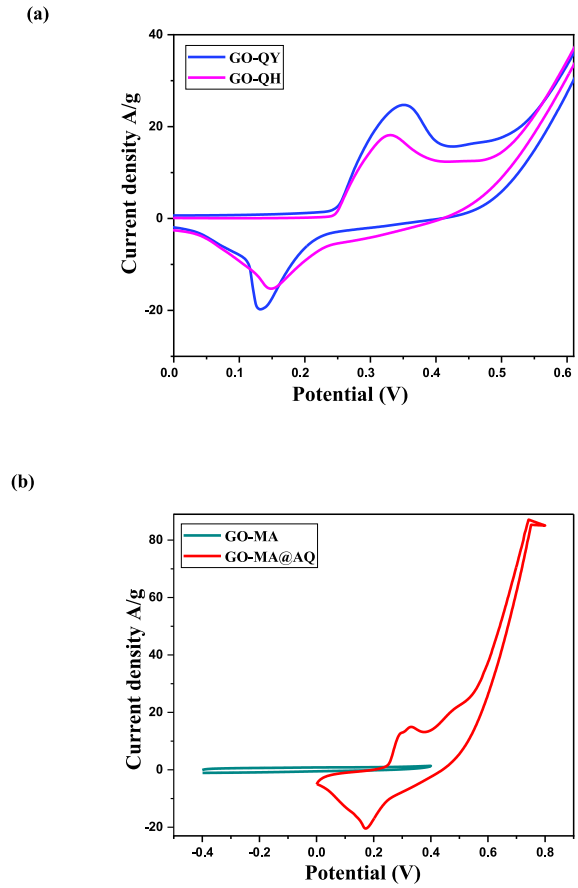


Fig. 4. Comparative Cyclic voltammograms (CV) of samples (a) GO-QH & GO-QY (b) GO-MA & GO-MA@AQ on Nickel foam working electrode at a scan rate of 10 mV s^{-1}.

reactions. This highlights the transition from reactions occurring in bulk to those occurring on the surface. This indicates that GO-MA@AQ primarily functions via surface kinetics rather than a bulk-controlled diffusion process, as the contribution from capacitive charge storage, as calculated, is 80 %.

The overall electrochemical performance analyzed from CV for both GO-QY and GO-MA@AQ, depicts the predominant contribution from surface-controlled process, indicating a capacitive charge storage mechanism across all scan rates. This can be attributed to the redox activity of quinone groups in both quinoline yellow and melamine-anthraquinone in facilitating the surface reactions at the electrode, thereby promoting the electrochemical performance. Also, the prominent increase in the pseudocapacitive behaviour of GO-MA@AQ as compared to GO-MA is ascertained to the electrochemical reversibility imposed by the heteroatom functionalization, resulting in better capacitive performance.

3.4.2. Galvanostatic charge-discharge studies

The quinone-N modified GO electrodes were characterized using the Galvanostatic charge-discharge technique to determine their capacitance performance and the results are plotted. The GCD analysis was performed in the charging potential of 0.4 V at various current densities (1–20 A g^{-1}) as shown in Fig. S6 (a-d). Obviously, the charging and discharging time of the modified GO electrodes are much longer with non-linear curves and plateaus, which is due to the combination of EDLC and faradaic capacitance. The GCD profiles of all the GO electrodes are compared, wherein GO-QY and GO-MA@AQ shows increased discharge time with more pronounced curve area during discharging process (Fig. 5(a) and (b)). This demonstrates superior capacitance indicating the material's high charge storage behaviour with significantly high discharge time. The specific capacitance (C_g) of the electrodes is calculated from the discharge curves using (1), where I is the current density, Δt the discharge time, m the mass of the active material, and ΔV the voltage drop during the discharge process. The specific capacitance of GO-QY was found to be 1052 F g^{-1} at 1 A g^{-1}. Similarly, GO-MA@AQ exhibited relatively higher specific capacitance 1853 F g^{-1} at a current density of 1 A g^{-1} as compared to GO-MA (Table S1).

$$C_s = \frac{I\,\Delta t}{m\,\Delta V} \tag{1}$$

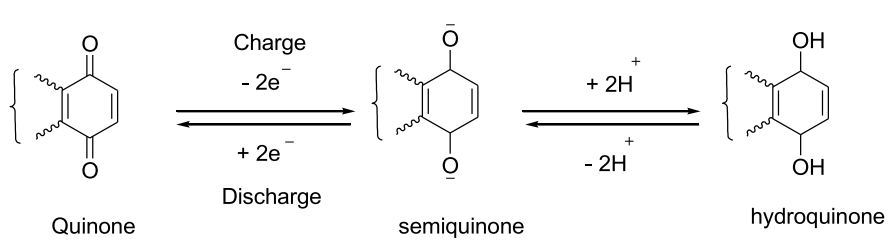
Consistent with the literature [42], specific capacitance decreased with increasing current densities Fig. S7, that is from 1 to 20 A g^{-1}, the specific capacity decreased from 1052 to 250 F g^{-1} for GO-QY. For GO-MA@AQ, C_s decreased from 1853 to 600 F g^{-1}. This is due to the diffusion limitations of K^+ ions within the electrode matrix. At lower current densities, the extended time available for ion diffusion allows K^+ ions to penetrate deep into the electrode material, facilitating interaction with a greater number of electroactive sites. This enables the utilization of the entire active surface area, leading to higher charge storage and increased specific capacitance. However, at higher current, the rapid potential sweep imposes time constraints on ion diffusion, restricting ion access to only the outermost electrode surface.

The energy and power densities are determined from the GCD curves using equations (2) and (3). Ragone plot was illustrated in Fig. 6 (a) which serves as a crucial tool for evaluating and comparing the performance of diverse energy storage devices, by illustrating the relationship between energy density and power density. Typically, energy storage devices that exhibit high energy density tend to demonstrate lower power density, and vice versa, discerning the suitability of the system that guides the choice of material and its design. Herein, GO-QY demonstrated high energy density of 85 W h kg^{-1}, with power density of 9.4 W kg^{-1}. Similarly, GO-MA@AQ has maximum energy density of 148 W h kg^{-1} and a power density of 14.4 W kg^{-1} which is higher than that of its counterpart material GO-MA (24 W h kg^{-1}).

$$E = 0.5C_s\,(\Delta V)^2 \tag{2}$$

$$P = 3.6\frac{E}{\Delta t} \tag{3}$$

The coulombic efficiency (%) of the fabricated GO electrodes is calculated using the equation, $\eta = \frac{Td}{Tc}\ x\ 100$, where T_d and T_c are



Scheme 3. Electron shift in Quinone moieties.

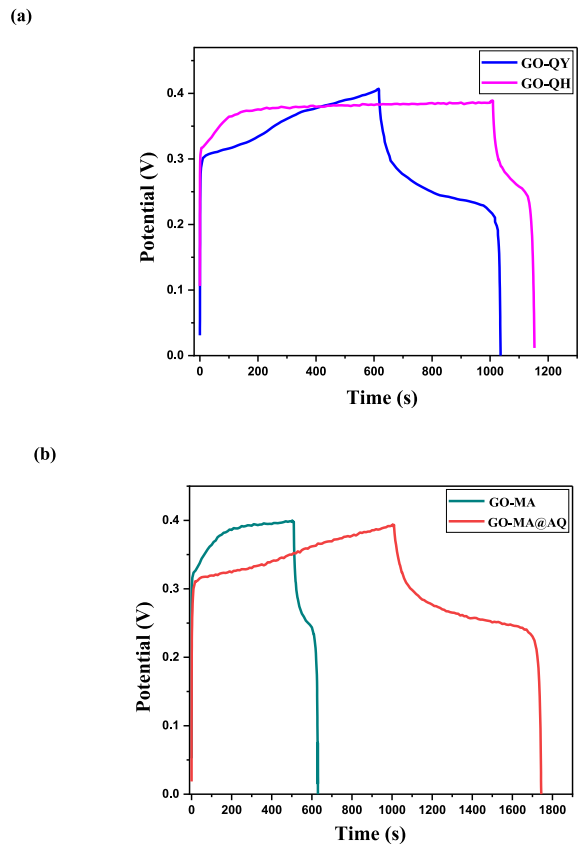


Fig. 5. Galvanostatic charge and discharge curves collected at a loading current of 1 A g^{-1} for (a) GO-QH & GO-QY (b) GO-MA & GO-MA@AQ

discharge and charge times, respectively. GO-QY exhibited a coulombic efficiency of 90 %, whereas GO-MA@AQ has a maximum efficiency of 95 % at 4 A g^{-1} (Fig. 6 (b)). The coulombic efficiency tends to decrease at lower current densities, which is indicative of the kinetic limitations in the charge storage process. Also, due to slower charge-discharge kinetics with increased ion diffusion resistance, the efficiency tends to decrease, often influenced by the intrinsic morphology and porosity of the electrode material. Furthermore, the cycling stability of the material was evaluated to assess its long-term durability. As shown in Fig. 6 (c), the stability test was conducted, wherein GO-MA@AQ showed maximum retention of 90 %, and GO-QY retained 86 % of its initial capacitance after 5000 continuous charge-discharge cycles (Table 2).

3.4.3. Electrochemical impedance spectroscopy

Electrochemical impedance tests were carried out with a frequency range from 100 kHz to 0.1 Hz to evaluate the charge transfer and electrolyte diffusion in GO-QY and GO-MA@AQ. The Nyquist plots presented in Fig. 7 illustrate the relationship between the real part (Z′) and the imaginary part (−Z″) of impedance for both GO-QY and GO-MA@AQ electrodes. The presence of a semicircular arc in the high-frequency region corresponds to the charge transfer resistance (R_{ct}), while the inclined line in the low-frequency region signifies Warburg-type diffusion characteristics. The impedance spectra were fitted using a modified Randle's equivalent circuit model, which effectively simulates the charge transport behavior at the electrode-electrolyte interface. The Nyquist plots of both the electrodes reveal significant differences in electrochemical behavior. The obtained solution resistance (R_s) values are 0.41 and 0.25 Ω for GO-QY and GO-MA@AQ, respectively and this indicates that GO-MA@AQ exhibits lower resistance to electrolyte solution. As shown in Fig. 7 (inset) GO-MA@AQ exhibits the lowest charge transfer resistance (R_{ct}), followed by GO-QY. This suggests that the inclusion of the triazine(C–N) unit along with AQ moiety enhances electrical conductivity and facilitates faster kinetics due to its redox active centers. Moreover, the Nyquist plot of GO-MA@AQ did not show any

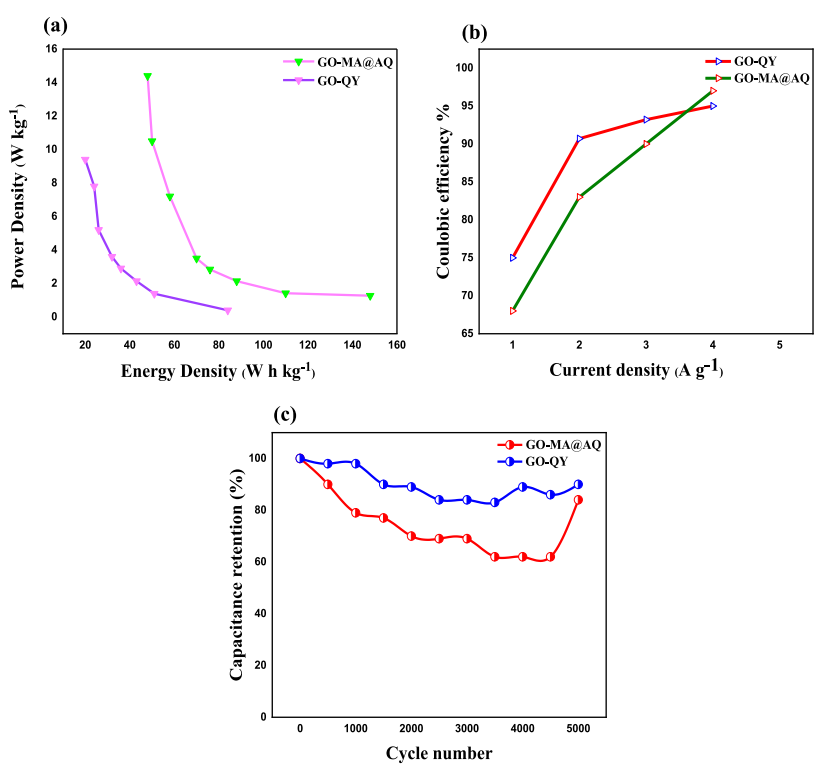


Fig. 6. (a) Ragone plot of GO-QY and GO-MA@AQ, (b) plot of current density vs. columbic efficiency, (c) plot showing capacitance retention vs. cycle number.

Table 2
Comparative study with previously reported Quinone derivatives.

Electrode Material	Synthesis methodology	Electrolyte	Electrochemical technique	Scan rate (m V s^{-1})	Current density (A g^{-1})	Specific capacitance (F g^{-1})	Energy Density (W h Kg^{-1})	Ref
AQ-HPCNTs	Carbonization-assisted solvothermal synthesis	1.0 M H_2SO_4	Galvanostatic charge discharge	–	1.0	710	–	[43]
DFGN	Hydrothermal synthesis	1.0 M H_2SO_4	Galvanostatic charge discharge	–	1.0	667.3	9.2	[44]
2-DCTQ-RGO	Sonochemical process	1.0 M H_2SO_4	Galvanostatic charge discharge	–	1.0	660	45	[45]
N-HtrGO/PQ	Hydrothermal synthesis	1.0 M H_2SO_4	Cyclic voltammetry	50.0	–	605	–	[46]
N-RGO@AAQ	Solvothermal synthesis	1.0 M H_2SO_4	Galvanostatic charge discharge	–	0.6	532.4	21.13	[47]
NRGO	Hydrothermal Method	6.0 M KOH	Galvanostatic charge discharge	–	0.1	179.8	–	[48]
GF-NG	Hydrothermal Method	6.0 M KOH	Galvanostatic charge discharge	–	0.6	380	15.2	[49]
N-rGOs	Hydrothermal route	6.0 M KOH	Cyclic voltammetry	100	–	244	–	[50]
$Ni(OH)_2$/ NCDs	Hydrothermal fabrication	6.0 M KOH	Galvanostatic charge discharge	–	1.0	1711	–	[51]
AQ-CDC	Electrochemical grafting	1.0 M KOH	Cyclic voltammetry	50.0	–	>300	–	[52]
AQ/AC	Sublimation	1.0 M H_2SO_4	Galvanostatic charge discharge	–	1.0	485	–	[53]
Substituted PAQs	Ring opening polymerization	10 M NaOH	Galvanostatic charge discharge	–	–	>700	143	[54]
Q_3DG	Chemical-reduction induced self-assembly	1.0 M Na_2SO_4	Cyclic voltammetry	10.0	–	386.3	–	[55]
NQrGO/CF	Hydrothermal self-assembly	6.0 M KOH	Galvanostatic charge discharge	–	0.2	223.8	50.7	[56]
GO-QH	Thermal and sonochemical-assisted method	6.0 M KOH	Galvanostatic charge discharge	–	1.0	370	29.6	This work
GO-MA	Thermal and sonochemical-assisted method	6.0 M KOH	Galvanostatic charge discharge	–	1.0	307	24.5	This work
GO-QY	Thermal and sonochemical-assisted method	6.0 M KOH	Galvanostatic charge discharge	–	1.0	1052	85	This work
GO-MA@AQ	Thermal and sonochemical-assisted method	6.0 M KOH	Galvanostatic charge discharge	–	1.0	1853	148	This work

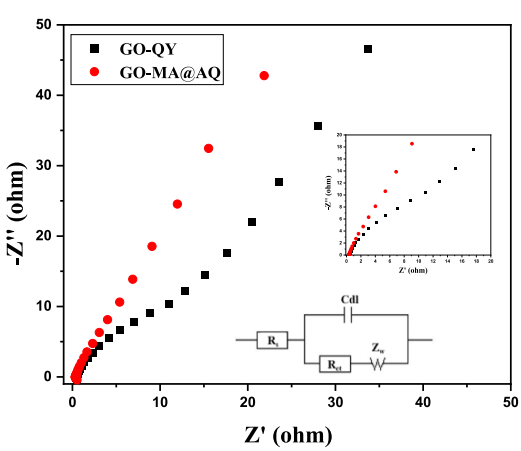


Fig. 7. Nyquist plots of GO-QY and GO-MA@AQ with the enlarged view of high-frequency region in the inset.

evident of semicircle indicating negligible charge transfer resistance demonstrating pronounced capacitive charge storage behavior. The linear curve inclined at 45° signifies ion diffusion resistance and from the Nyquist plot, the more vertical slope in the low frequency region for both the electrodes indicates lower ion diffusion resistance. Further, the inclined line of GO-MA@AQ is more aligned to 45° than GO-QY, representing rapid ion transfer. The Nyquist plots were interpreted using a Randles equivalent circuit model, comprising solution resistance (R_s), Warburg impedance (Z_w), charge transfer resistance (R_{ct}), and double layer capacitance (C_{dl}) as shown in the inset of Fig. 7. These improvements can be ascribed to interaction between functionalized organic moieties and the GO matrix, contributing to superior electrochemical performance.

4. Conclusion

This study introduces a novel design strategy for enhancing the electrochemical performance of graphene oxide through synergistic dual functionalization with quinone units and nitrogen-rich organic moiety. The successful integration of redox-active quinone and electron-donating nitrogen functionalities into the GO framework was confirmed via comprehensive physicochemical characterizations, validating the structural and chemical modification of the GO matrix. Electrochemical evaluations revealed a remarkable enhancement in specific capacitance across all functionalized GO derivatives when compared to pristine GO. Among them, GO-QY exhibited a significantly improved specific capacitance of 1052 F g^{-1}, far surpassing the performance of GO-QH (370 F g^{-1}), highlighting the critical role of molecular structure in redox activity. Notably, the melamine-anthraquinone co-functionalized composite, GO-MA@AQ, delivered an exceptional specific capacitance of 1853 F g^{-1}, representing over a sixfold enhancement relative to pristine GO. The findings demonstrated a marked increase in specific capacitance for melamine-anthraquinone functionalized GO composite, surpassing that of the unmodified GO system, reaffirming the effectiveness of heteroatom functionalization in optimizing electrochemical behavior. The synergy between redox-active functionalities and nitrogen heteroatoms effectively promotes charge transfer, boosts conductivity, and stabilizes the electrochemical response. These findings highlight the effectiveness of targeted organic functionalization in tailoring the electronic and structural properties of GO-based materials, paving the way for the development of environmentally friendly, cost

effective, and energy-efficient electrodes for next-generation supercapacitor systems, marking a significant advancement in the design of high-performance carbonaceous supercapacitor electrodes.

CRediT authorship contribution statement

K. Keerthana: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **M. Nigotha:** Writing – original draft, Methodology, Investigation, Formal analysis. **A. John Maria Xavier:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Conceptualization.

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Not Applicable.

Declaration of competing interest

There are no conflict of interest to declare.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.matchemphys.2025.131511.

Data availability

Data will be made available on request.

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